Mathematical and Computational Methods

Exercise Sheet 7: Differentiation I: limits and the rules

This sheet accompanies *Lecture 7 (Differentiation I: limits and the rules)*. It exercises every skill of the unit: evaluating limits (including at infinity), continuity, the derivative from first principles, the standard derivatives and the sum, product, quotient and chain rules, and finally parametric, implicit and logarithmic differentiation. Work in radians throughout.

How to use this sheet. Problems are graded by difficulty—⋆ drill (routine fluency), ⋆⋆ core (standard, tutorial level) and ⋆⋆⋆ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ▶) and **further practice** for independent home study.

Solutions. This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showsolutions{}\input{MCM_Exercises_Unit7}"`) and recompile.

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1 Warm-up drills

Limits, continuity

Exercise 1.1 *(⋆ drill limits at a point)*. Evaluate, cancelling a common factor or rationalising first where needed:

(a) $\displaystyle\lim_{x\to 2}\frac{x^2-4}{x-2}$;

(b) $\displaystyle\lim_{x\to 3}\frac{x^2-9}{x^2-x-6}$;

(c) $\displaystyle\lim_{x\to 0}\frac{\sqrt{x+1}-1}{x}$;

(d) $\displaystyle\lim_{x\to 0}\frac{\sin 5x}{x}$.

Exercise 1.2 *(⋆ drill one-sided and infinite limits)*. (a) Find $\displaystyle\lim_{x\to 0^+}\frac{1}{x}$ and $\displaystyle\lim_{x\to 0^-}\frac{1}{x}$, and state the vertical asymptote.

(b) For $f(x)=\begin{cases}x^2, & x<1\\ 2x, & x\ge 1\end{cases}$ find the one-sided limits at $x=1$ and decide whether $\lim_{x\to 1} f(x)$ exists.

Exercise 1.3 *(⋆ drill limits at infinity / horizontal asymptotes)*. Divide by the highest power and state any horizontal asymptote $y=L$:

(a) $\displaystyle\lim_{x\to\infty}\frac{3x^2-1}{x^2+5}$;

(b) $\displaystyle\lim_{x\to\infty} \frac{2x+1}{x^2+3}$;

(c) $\displaystyle\lim_{x\to\infty} \frac{4x^3-x}{2x^3+7}$.

Exercise 1.4 *(⋆ drill continuity)*. Find the constant that makes each function continuous everywhere.

(a) $f(x) = \dfrac{x^2-1}{x-1}$ for $x \neq 1$, and $f(1) = k$.

(b) $g(x) = \begin{cases} cx+1, & x \leq 2 \\ x^2, & x > 2. \end{cases}$

Exercise 1.5 *(⋆ drill asymptotes and a sketch)*. For $f(x) = \dfrac{2x}{x-1}$:

(a) find the vertical asymptote (where the denominator vanishes) and the horizontal asymptote from $\lim_{x\to\pm\infty} f(x)$;

(b) make a rough sketch marking *both* asymptotes, the intercept at the origin, and the two branches.

The derivative; standard derivatives and rules

Exercise 1.6 *(⋆ drill first principles)*. Using $f'(x) = \lim_{\delta x\to 0} \dfrac{f(x+\delta x)-f(x)}{\delta x}$, differentiate from first principles: (a) $f(x) = x^2 - 3x$; (b) $f(x) = \dfrac{1}{x}$.

Exercise 1.7 *(⋆ drill standard derivatives)*. Write down the derivative of each (use a power where helpful):

(a) x^5, (b) $\sqrt{x}$, (c) $\dfrac{1}{x^3}$, (d) e^{2x}, (e) $\ln x$, (f) $\sin 3x$, (g) $\cos\frac{x}{2}$, (h) $\tan x$, (i) $\cosh x$.

Exercise 1.8 *(⋆ drill sum / constant-multiple)*. Differentiate: (a) $3x^4 - 2x^2 + 7x - 5$; (b) $\dfrac{2}{x} + 3\sqrt{x}$.

Exercise 1.9 *(⋆ drill product rule)*. Use $(uv)' = u'v + uv'$: (a) $x^2\mathrm{e}^x$; (b) $x\sin x$.

Exercise 1.10 *(⋆ drill quotient rule)*. Use $\left(\frac{u}{v}\right)' = \dfrac{u'v-uv'}{v^2}$: (a) $\dfrac{x}{x+1}$; (b) $\dfrac{\sin x}{x}$.

Exercise 1.11 *(⋆ drill chain rule)*. Differentiate: (a) $(3x+1)^5$; (b) e^{x^2}; (c) $\sin(x^2)$; (d) $\ln(1+x^2)$.

Further techniques

Exercise 1.12 *(⋆⋆ core parametric, implicit, logarithmic drill)*. (a) Parametric: find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for (i) $x = t^2$, $y = t^3$; (ii) $x = \cos t$, $y = \sin t$.

(b) Implicit: find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for (i) $x^2 + y^2 = 25$; (ii) $xy = 1$.

(c) Logarithmic: find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for (i) $y = 2^x$; (ii) $y = x^x$.

2 Tutorial problems

Problems marked ▶ form the live ∼2-hour tutorial core; the rest are for completion at home.

Exercise 2.1 *(▶ ⋆⋆ core a limit medley + concept)*. Evaluate, naming the technique each time:

$$\text{(a)}\ \lim_{x\to 0}\frac{\sin x}{x},\quad \text{(b)}\ \lim_{x\to\infty}\frac{4x^2-3x+1}{2x^2+5},\quad \text{(c)}\ \lim_{x\to 3}\frac{x^2-9}{x-3},\quad \text{(d)}\ \lim_{x\to 0^\pm}\frac{1}{x}.$$

Concept. In (c) the function is undefined at $x=3$, yet the limit exists. Explain in one sentence why this is no contradiction.

Exercise 2.2 *(▶ ⋆⋆ core continuity and differentiability)*. Let $f(x)=\begin{cases} x^2, & x\le 1\\ ax+b, & x>1.\end{cases}$ Find a,b so that f is *differentiable* at $x=1$. Why is your f automatically continuous there?

Exercise 2.3 *(▶ ⋆⋆ core first principles and the tangent (geometric reading))*. Let $f(x)=\sqrt{x}$.

(a) Show from first principles that $f'(x)=\dfrac{1}{2\sqrt{x}}$.

(b) Hence find the equation of the tangent line to $y=\sqrt{x}$ at $x=4$.

Exercise 2.4 *(⋆⋆ core velocity and acceleration — physics)*. A particle moves along a line with displacement $s(t)=t^3-6t^2+9t$ metres, $t\ge 0$ seconds.

(a) Find the velocity $v(t)=\dot{s}$ and acceleration $a(t)=\ddot{s}$.

(b) When is the particle momentarily at rest?

(c) In which direction is it moving at $t=2$?

(d) Give the acceleration at each rest instant and interpret its sign.

Exercise 2.5 *(⋆⋆ core sensitivity and the sigmoid — data science)*. (a) The logistic function is $\sigma(x)=\dfrac{1}{1+e^{-x}}$. Show, using the chain rule, that $\sigma'(x)=\sigma(x)\big(1-\sigma(x)\big)$.

(b) A model's error is $L(w)=(w-3)^2+1$. Compute $\frac{dL}{dw}$ and evaluate it at $w=5$. Reading the derivative as a *sensitivity*, should a gradient-descent step increase or decrease w? Write the update with learning rate η.

Exercise 2.6 *(▶ ⋆⋆ core product and quotient together)*. Differentiate and simplify: (a) $y=(x^2+1)(x^3-x)$; (b) $y=\dfrac{e^x}{x^2+1}$.

Exercise 2.7 *(▶ ⋆⋆ core quotient-then-chain (worked-example anchor))*. Differentiate $y=\dfrac{(2x-1)^4}{x^2+1}$, simplifying as far as possible.

Exercise 2.8 *(⋆⋆ core nested chain rule)*. Differentiate: (a) $y=\ln\big(\sqrt{x^2+1}\big)$; (b) $y=\sqrt{1+\sqrt{x}}$.

Exercise 2.9 *(▶ ⋆⋆ core implicit differentiation (worked-example anchor))*. The folium of Descartes is $x^3+y^3=6xy$.

(a) Find $\frac{dy}{dx}$ by implicit differentiation.

(b) Hence find the slope of the curve at the point $(3,3)$.

Exercise 2.10 *(⋆⋆ core an implicit second derivative)***.** For the hyperbola $x^2 - y^2 = 16$, find $\frac{dy}{dx}$ and then $\frac{d^2y}{dx^2}$, expressing the second derivative purely in terms of y.

Exercise 2.11 *(▶ ⋆⋆ core parametric tangents)***.** A curve is given by $x = t^2,\ y = t^3 - 3t$.

(a) Find $\frac{dy}{dx}$ in terms of t.

(b) Find the point(s) with a horizontal tangent and the point(s) with a vertical tangent.

Exercise 2.12 *(⋆⋆ core logarithmic differentiation — a multi-factor expression)***.** Use logarithmic differentiation to differentiate $y = (x^2+1)^3(2x-1)^4\,e^{-5x}$.

Exercise 2.13 *(⋆⋆ core spot the error —* conceptual*)***.** Each line contains one mistake. Identify it and give the correct derivative.

(a) "$\frac{d}{dx}(x^2+1)^3 = 3(x^2+1)^2$."

(b) "$\frac{d}{dx}\left(x^2 e^x\right) = 2x\,e^x$."

(c) "$\frac{d}{dx}\left(\sin x \cos x\right) = \cos x\,(-\sin x)$."

3 Further practice (home)

Exercise 3.1 *(⋆ drill more limits)***.** Evaluate:

$$\text{(a)}\ \lim_{x\to 4}\frac{x^2-16}{x-4},\quad \text{(b)}\ \lim_{x\to\infty}\frac{3x^3-2x}{6x^3+1},\quad \text{(c)}\ \lim_{x\to\infty}\frac{x^2+1}{x^3-1},$$

$$\text{(d)}\ \lim_{x\to\infty}\frac{x^3}{x^2+1},\quad \text{(e)}\ \lim_{x\to 1}\frac{x-1}{\sqrt{x}-1},\quad \text{(f)}\ \lim_{x\to 0}\frac{\sin 2x}{3x}.$$

Exercise 3.2 *(⋆⋆ core classify the discontinuities)***.** For each function, find where it fails to be continuous and classify the break as *removable*, a *jump*, or *infinite*.

$$\text{(a)}\ f(x) = \frac{x-2}{x^2-4},\qquad \text{(b)}\ g(x) = \frac{1}{x-1},\qquad \text{(c)}\ h(x) = \begin{cases} x, & x<0 \\ x+2, & x\ge 0. \end{cases}$$

Exercise 3.3 *(⋆⋆ core first principles, cubic)***.** Show from first principles that if $f(x) = x^3$ then $f'(x) = 3x^2$.

Exercise 3.4 *(⋆⋆ core higher-order derivatives)***.** (a) Find a formula for the nth derivative of (i) $y = e^{2x}$; (ii) $y = \dfrac{1}{x}$.

(b) For $y = x^4$, list $y', y'', y''', y^{(4)}, y^{(5)}$.

(c) Find the nth derivative of $y = \sin x$.

Exercise 3.5 *(⋆⋆ core the differentiation bank)***.** Differentiate (state the rule used):

$$\text{(a)}\ e^{x^2}\sin x,\quad \text{(b)}\ x^2\ln x,\quad \text{(c)}\ \frac{x^2+1}{x^2-1},\quad \text{(d)}\ e^{-x}\cos 2x,\quad \text{(e)}\ \frac{\ln x}{x},\quad \text{(f)}\ \cos^3 x.$$

Exercise 3.6 *(⋆⋆ core parametric bank, including a second derivative)***.** (a) For $x = 2t,\ y = t^2 - 1$ find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

(b) For the circle $x = r\cos t$, $y = r\sin t$ find $\frac{\mathrm{d}y}{\mathrm{d}x}$ and $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$.

Exercise 3.7 *(⋆⋆ core implicit bank, including a tangent line)*. (a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for $x^2 - xy + y^2 = 7$.

(b) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for $\sin(xy) = x$.

(c) Find the equation of the tangent to $x^2 + y^2 = 25$ at the point $(3, 4)$.

Exercise 3.8 *(⋆⋆ core logarithmic bank)*. Use logarithmic differentiation:

(a) $y = (x+1)^3(2x-1)^4(x^2+1)^2$;

(b) $y = \dfrac{\sqrt{x^2+1}\,\mathrm{e}^x}{(x+3)^4}$.

Exercise 3.9 *(⋆⋆ core simple harmonic motion — physics)*. A mass oscillates with $x(t) = A\cos\omega t$ (A, ω constant).

(a) Find $\dot{x}$ and $\ddot{x}$.

(b) Show that $\ddot{x} = -\omega^2 x$, i.e. the acceleration is proportional to the displacement and oppositely directed.

Exercise 3.10 *(⋆⋆ core activation and a learning curve — data science)*. (a) The *softplus* activation is $f(x) = \ln(1+\mathrm{e}^x)$. Using the chain rule, show that $f'(x) = \dfrac{1}{1+\mathrm{e}^{-x}}$ — i.e. its slope is exactly the logistic sigmoid.

(b) During training a loss decays as $L(t) = \mathrm{e}^{-t/5}$, where t is the epoch number. Find the instantaneous rate of change $L'(t)$, and evaluate it at $t = 0$ and $t = 5$.

Exercise 3.11 *(⋆⋆ core true or false — conceptual)*. State whether each is true or false, with a one-line reason.

(i) If f is continuous at a then f is differentiable at a.

(ii) $(uv)' = u'v'$.

(iii) For $\lim_{x\to a} f(x)$ to exist, $f(a)$ must be defined.

(iv) If $\lim_{x\to\infty} f(x) = L$ then $y = L$ is a horizontal asymptote.

(v) $\frac{\mathrm{d}}{\mathrm{d}x}x^n = nx^{n-1}$ holds only for integer n.

(vi) $f(x) = x^{1/3}$ is differentiable at $x = 0$.

Exercise 3.12 *(⋆⋆⋆ challenge harder limits)*. Evaluate, quoting $\lim_{u\to 0}\frac{\sin u}{u} = 1$ where needed:

$$\text{(a)}\ \lim_{x\to 0}\frac{1-\cos x}{x^2}, \quad \text{(b)}\ \lim_{x\to\infty}\left(\sqrt{x^2+x} - x\right), \quad \text{(c)}\ \lim_{x\to 0}\frac{\sin 3x}{\sin 5x}.$$

Exercise 3.13 *(⋆⋆⋆ challenge the cycloid)*. A cycloid is traced by $x = a(t - \sin t)$, $y = a(1-\cos t)$ ($a > 0$). Show that $\frac{\mathrm{d}y}{\mathrm{d}x} = \cot\frac{t}{2}$, and state the values of t at which the tangent is horizontal.

Exercise 3.14 *(⋆⋆⋆ challenge a variable power)*. Use logarithmic differentiation to differentiate $y = x^{\sin x}$ for $x > 0$.

Exercise 3.15 *(⋆⋆⋆ challenge a tower of powers)*. Differentiate $y = x^{x^x}$ for $x > 0$. *Hint: take logs twice, or differentiate x^x first.*

Exercise 3.16 *($\star\star\star$ challenge a smooth join (extension))*. Find constants a, b, c so that $f(x) = \begin{cases} x^3, & x \leq 1 \\ ax^2 + bx + c, & x > 1 \end{cases}$ is *twice* differentiable at $x = 1$.